FinTwin: An Overhead-Aware Financial Digital Twin for Business Investment Decisions

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Group 6 · Engineering Economics and Financial Decision Making · CIE Activity 2 Mini Project
Live application: https://fintwin-0ae4.onrender.com · Code: https://github.com/shreyank26pal-dev/eefm_project

Abstract — Our CIE-1 survey of eleven IEEE/ACM studies justified a Financial Digital Twin (FDT) with a deterministic NPV of $684,814 (IRR 60.3%). The same literature warns that a twin's computational, energy and integration overheads can erode the value it creates (Gap 2), and that most results are validated only in simulation (Gap 4). This project builds **FinTwin**, a cloud-deployed Flask application. It mirrors a manufacturing SME day by day, maps its physical state to financial state, and prices the twin itself with an overhead-aware Monte-Carlo capital-budgeting model discounted at a volatility-adjusted rate $r(\sigma)$. The failure model is trained on the real AI4I 2020 dataset (ROC-AUC 0.963). FinTwin reproduces the survey case to within $67. Once hidden overheads and risk are priced, the expected NPV falls 45%, to $375k. A high-fidelity twin that looks acceptable statically (+$243k) is rejected (−$273k, with a 92.5% probability of loss). With benefits measured by the twin instead of assumed, only a Lite twin keeps a positive expected NPV.

Keywords — Financial digital twin, Monte-Carlo NPV, risk-adjusted discount rate, condition-based replacement, EUAC, cash conversion cycle, predictive maintenance, decision support system.

I. Introduction and Problem Statement

The CIE-1 survey showed that AI digital twins can turn engineering economics from a quarterly, retrospective calculation into a real-time control loop. Live telemetry compresses the cash conversion cycle (CCC), feeds dynamic credit scoring and drives condition-based replacement [A5, A9, A11]. Its Section 5 case study valued a $450,000 twin for a $35M automotive supplier at NPV = $684,814, IRR = 60.3% and a payback of 1.5 years.

That valuation has three weaknesses the survey itself exposed:

1. **It is deterministic.** The benefits B1–B3 are fixed point estimates and the discount rate is a flat 10%, although the survey's own Eq. 4–5 call for a stochastic NPV with a volatility-adjusted rate.
2. **It ignores the twin's own running costs.** Yoon et al. [A7] warn that energy, hardware replacement and computation can erode the savings (Gap 2). The case study books energy inside a flat $45k OPEX and has no hardware refresh, no integration ramp-up and no compliance cost.
3. **Its benefits are assumed, not measured.** The 32-day CCC reduction and the 3.27-point spread cut are borrowed from different papers and studies (Gap 4).

Problem statement. Build and evaluate a web-based, AI-enabled Financial Digital Twin that (a) simulates how a business's operations become financial outcomes with and without a twin; (b) measures the twin's benefits instead of assuming them; and (c) decides whether a twin is worth building, and at what fidelity, using risk-adjusted, overhead-aware capital budgeting.

II. Literature Gap Identification

FinTwin extends one framework from the survey: the Financial Digital Twin of Vaghani et al. [A9], which maps material flows onto accounts receivable, accounts payable and holding costs. It adds the valuation, the overhead accounting and the data validation that the reviewed papers omit (Table I).

TABLE I. Survey gaps and FinTwin's response

Gap / limitation	Source in survey	FinTwin response
Static, deterministic capital budgeting	Sec. 5 case study; Eq. 3	Monte-Carlo NPV/IRR with risk on realisation, growth and shocks; discounting at $r(\sigma)$ (Eq. 5)
Gap 2: overhead vs. net value	[A7]; 66 W edge nodes [A1]; ZKP engines [A10]	Energy, refresh, compliance, ramp-up and OPEX escalation priced; Lite / Standard / High tiers compared
Gap 4: simulation-only validation	[A1], [A10], [A11]	Failure model trained on real AI4I 2020 data; benefits measured by a two-world twin
Replacement stated, not operationalised	Eq. 6–7; Jidoka twin [A5]	ML P_{fail} drives Eq. 7 daily; derating implements the control action a_t of Eq. 4
Credit–telemetry link only described	[A11]; Eq. 8–9	Real-time credit score from on-time delivery, fleet health and collection efficiency
Heavy, SME-inaccessible architectures	[A1], [A3], [A7]	Lightweight web DSS on a free cloud tier, with an offline rule-based advisor

III. Proposed Methodology and Architecture

FinTwin has four layers (Fig. 1):

- **Physical / transaction layer:** six CNC spindles with AI4I-style telemetry, inventory, payables and customer invoices.
- **AI twin layer:** a failure-probability model, an invoice-delay model, and the Eq. 7 replacement and derating logic.
- **Financial layer:** DIO, DSO, DPO, CCC, working capital, credit score and loan rate.
- **Decision layer:** Monte-Carlo valuation, tier selection and the AI advisor.

Every scenario runs twice on common random numbers, so the difference between the two worlds is the twin's measured benefit:

- **Baseline:** a static ERP view; time-based replacement at the OEM interval of 200 tool-wear minutes; a lender that sees only the historical credit score $C_{hist} = 0.30$.
- **Twin:** ML condition-based replacement and derating; delivery verified on 90% of invoices; a real-time composite credit score.

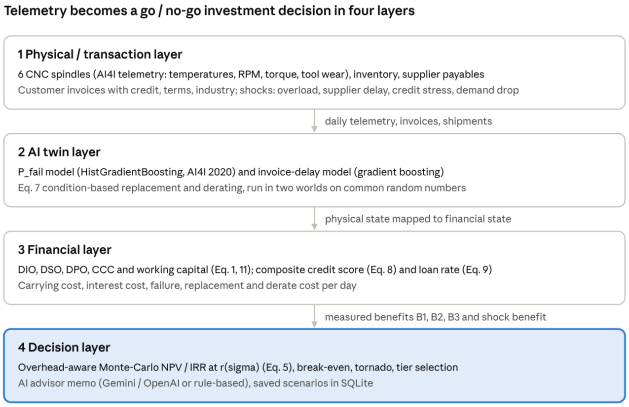


Fig. 1. FinTwin architecture: telemetry becomes a go/no-go investment decision in four layers.

The method has four steps:

1. Simulate 365 days for the survey's \$35M case firm, under steady and stressed conditions, with 5 seeds each.
2. Annualise the cost differences into three benefits: B1 (working-capital carrying cost), B2 (interest on a \$2.5M credit line) and B3 (failure, replacement and derating cost). The shock-year benefit is the stressed total minus the steady total.
3. Feed the survey-assumed or twin-measured benefits into the Monte-Carlo valuation for three twin tiers.
4. Recommend the tier with the highest risk-adjusted NPV whose probability of loss is at most 10%.

Data. The failure model uses the AI4I 2020 Predictive Maintenance dataset (UCI #601, CC BY 4.0) [12]: 10,000 records with 3.39% failures. Receivables ledgers are confidential, so a 6,000-invoice synthetic ledger encodes stylised facts: verified delivery shortens settlement [A9, A11], while weak credit, large invoices and long terms lengthen it.

IV. Economic Model and Decision Logic

A. Working capital (Eq. 1, 11). DIO follows an AR(1) path toward its target: 48 days in the baseline, 34 with the twin. The twin corrects 35% of a deviation per day; the baseline corrects 8%. DSO is the 30-day rolling mean of ML-predicted days-to-pay, and DPO is 35 days. B1 is the difference in carrying cost at 6%.

$$CCC_t = DIO_t + DSO_t - DPO_t$$
$$WC_t = \tfrac{1}{365}\left(DIO_t\, COGS + DSO_t\, Rev - DPO_t\, COG\right.$$

B. Real-time credit (Eq. 8–9). The real-time score combines on-time delivery OTD, fleet failure rate FR and collection efficiency CE. Without a twin, the lender prices C_{hist} only. The parameters are $\alpha = 0.4$, $r_{base} = 8\%$, $\Delta r_{max} = 15\%$ and $k = 3$.

$$C_t^{rt} = 0.4\,OTD_t + 0.3\,(1 - 10\,FR_t) + 0.3\,CE_t$$
$$C_t = \alpha C_{hist} + (1-\alpha)\,C_t^{rt}$$
$$r_{loan} = r_{base} + \Delta r_{max}\, e^{-kC_t}$$

C. Condition-based replacement and derating (Eq. 4, 6–7). For a consumable tool, salvage is about zero, so the ΔS and rS terms of Eq. 7 vanish. Each day, the twin replaces a tool when its next-period marginal cost exceeds the challenger's EUAC:

- P_{fail} is the calibrated ML probability at the next day's wear. It is constrained to be non-decreasing in wear, a physics prior.
- The challenger's cost uses the mean hazard of a fresh tool and capital recovery of $C_{rep} = \$1{,}200$ over its design life L, with $C_{cat} = \$9{,}000$.

$$E[O\&M_{t+1}] + P_{fail}(D_{t+1})\,C_{cat} > C_{rep}(A/P, i, L) + E[O\&M_{new}] + \bar{P}_{fail}^{new}\,C_{cat}$$

Under overload, the twin can also derate a machine. It does so when the expected failure cost avoided exceeds the lost contribution margin (20% of output × 30% margin). This is the control action a_t of the Bellman form, Eq. 4.

D. Overhead-aware cash flow (Gap 2).

$$CF_y = \rho_y (1+g)^{y-1} \textstyle\sum_k \theta_k B_k\, \varepsilon_y + s_y\, \kappa\, B_{shock} - E_y - O_y - Comp - R_y$$

The terms in this equation are:

- ρ_y: integration ramp-up, 60% in year 1 and 90% in year 2.
- g: revenue growth, $\sim N(4\%, 3\%)$.
- θ_k: triangular benefit-realisation factors $(0.55, 0.90, 1.15)$.
- ε_y: yearly noise.
- s_y: a disruption year, $\sim$ Bernoulli(0.3), with severity $\kappa \sim U(0.5, 1.5)$.
- E_y: energy cost, nodes × watts × 8,760 h × PUE 1.5 × \$0.12/kWh, escalating 5% a year.
- O_y: fixed OPEX, escalating 3% a year.
- $Comp$: compliance, 4% of capex a year.
- R_y: hardware refresh, 30–35% of hardware cost in year 3.

E. Stochastic NPV and $r(\sigma)$ (Eq. 5). The cash flow is simulated 5,000 times. σ is the mean coefficient of variation of annual cash flow. With $r_f = 6.5\%$, $r_m = 13.5\%$ and $\beta = 0.5$, the WACC is exactly the survey's 10%; $\gamma = 0.10$.

$$r(\sigma) = r_f + \beta(r_m - r_f) + \gamma\sigma$$
$$NPV^{(j)} = \sum_{y=1}^{T} \frac{CF_y^{(j)}}{(1 + r(\sigma))^y} - I_0$$

The model reports:

- mean, P5 and P95 NPV;
- $P(NPV < 0)$ and CVaR at 5%;
- median IRR, by vectorised bisection;
- discounted payback;
- AW and FW, via (A/P) and (F/P);
- the break-even benefit-realisation factor;
- a ±20% tornado sensitivity chart.

F. Tier decision rule. Table II defines three twin tiers. FinTwin recommends the tier with the highest mean risk-adjusted NPV among those with $P(\text{loss}) \leq 10\%$. If none qualifies, it recommends the least-bad tier and advises a stage-gated pilot. A *decision reversal* is flagged when the static NPV is positive but the risk-adjusted NPV is negative.

TABLE II. TWIN FIDELITY TIERS

Tier	Capex	Fixed OPEX	Nodes × W	Capture B1/B2/B3/shock
Lite	$150k	$20k	6 × 15	0.55/0.40/0.30/0.35
Standard (survey)	$450k	$42.9k	20 × 66 [A1]	1/1/1/1
High (6G, GNN, ZKP)	$900k	$95k	60 × 120 [A1]	1.15/1.20/1.25/1.40

V. AI / WEB APPLICATION

The application is a Flask REST back end (`backend/`) with a single-page dashboard (`frontend/`, Fig. 2), about 2,300 lines of Python and JavaScript. The datasets and their generator live in `dataset/`. Its components are:

- **Economics engine:** `backend/fdt/economics.py` and `valuation.py` (NumPy, with vectorised Monte Carlo and IRR).
- **Twin simulator:** `backend/fdt/twin.py`, with batched ML inference over a wear grid.
- **ML models:** `backend/fdt/ml.py` (scikit-learn).
- **AI advisor:** `backend/fdt/advisor.py`. It uses Gemini or OpenAI when an API key is set, and a rule-based expert system otherwise. It receives only computed numbers as JSON and is told never to invent figures.
- **Front end and storage:** HTML/CSS/JS with Chart.js in `frontend/`, SQLite for saved scenarios, and Gunicorn as the server.

The dashboard has eight modules:

1. **Overview:** a one-click full pipeline.
2. **Live Twin:** a shock builder; CCC, loan-rate, cost and P_{fail} charts; the Eq. 7 decision log.
3. **Stochastic NPV:** histogram, static-to-risk bridge, cash-flow fan and tornado chart.
4. **Right-size Twin:** tier comparison with a decision-reversal banner.
5. **Replacement:** the Eq. 6 EUAC curve and the Eq. 7 decision, with and without the AI hazard term.
6. **TVM Toolkit:** PW, FW, AW, IRR, interest factors, NPV profile, and SL/DDB depreciation.
7. **AI Advisor:** a generated investment memo and Q&A chat.
8. **Scenarios:** saved runs.

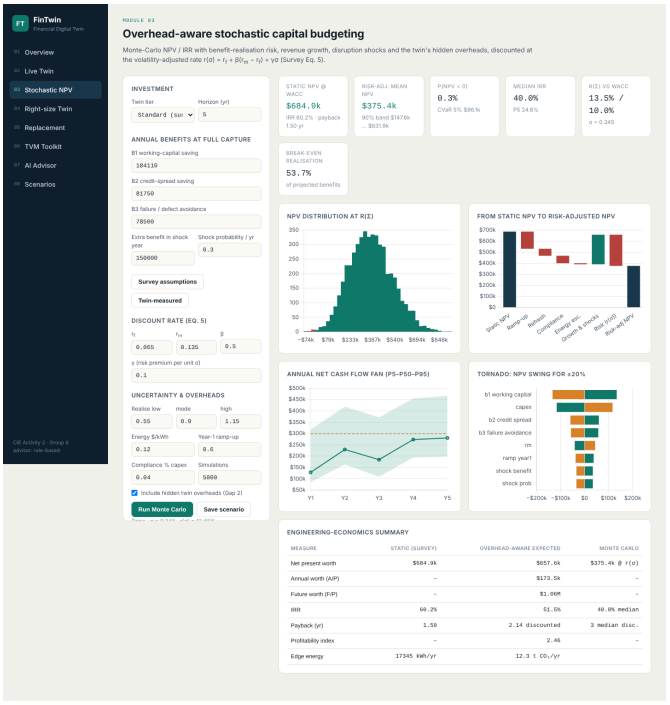


Fig. 2. Stochastic NPV module of the deployed application.

VI. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

Pricing risk and the twin's own overheads removes 45% of the survey NPV, and the benefits measured by the twin are about half of those the survey assumed. All results are reproducible with `python -m experiments.run_experiments` (fixed seeds), and 9 unit tests cover the economics engine.

E1 — Survey replication and revaluation (Table III, Fig. 3).

TABLE III. STANDARD TWIN VALUED FOUR WAYS

Method	NPV	IRR	Payback
Survey CIE-1 (reported)	$684,814	60.3%	1.50 yr
FinTwin static, same inputs	$684,881	60.2%	1.50 yr
+ overheads, expected @ 10%	$657,649	51.5%	2.14 yr disc.
+ Monte Carlo @ $r(\sigma)$=13.5%	$375,387	40.0% med.	3 yr disc. med.

Cash-flow volatility of $\sigma = 0.345$ raises the hurdle rate by 3.5 points. The project still has only a 0.28% chance of loss:

- the 90% NPV band is $148k–$632k, and CVaR(5%) is $96k;
- it breaks even if 53.7% of the assumed benefits materialise.

The present worth of the overheads is $155k for ramp-up loss, $68k for compliance, $63k for the hardware refresh and $9k for energy. Edge energy, the overhead the literature emphasises, is the smallest of the four.

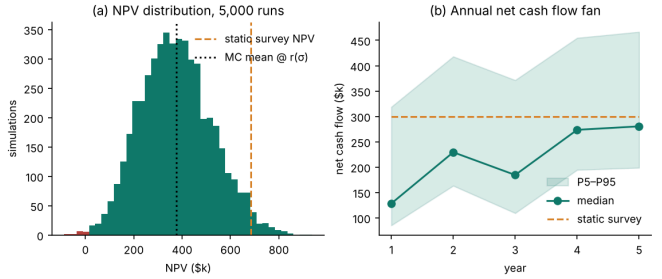


Fig. 3. (a) NPV distribution over 5,000 runs vs. the static survey NPV; (b) annual net cash flow fan (P5–P95) vs. the flat survey flow.

E2 — Measured vs. assumed benefits (Table IV, Fig. 4). Each run covers 365 days with 5 seeds. The stress scenario combines overload, supplier delay and customer credit stress.

TABLE IV. ANNUAL BENEFITS, $K (MEAN ± S.D.)

Benefit	Survey	Twin steady	Twin stress
B1 working capital	184.1	84.4 ± 1.9	92.6 ± 2.0
B2 credit spread	81.8	93.8 ± 0.9	91.1 ± 0.6
B3 failure avoidance	78.5	4.9 ± 16.0	197.5 ± 47.8
Total	**344.4**	**183.1 ± 16.0**	**381.2 ± 46.5**

The twin cuts the CCC from 55.1 to 36.3 days and the loan rate from 14.10% to 10.35%. Even so, B1 is under half the survey figure, because ML-predicted DSO falls by 4.8 days, not 18.

In steady state, the Eq. 7 rule only matches a well-chosen fixed interval. Under stress, derating cuts failures from 68.2 to 39.0 a year, so the twin's value is concentrated in disruption years. Re-valued with the measured benefits (an extra $198k in a shock year at $p = 0.3$), the Standard twin has a static NPV of +$73.6k but a risk-adjusted mean of −$73.3k, with a 71.2% probability of loss.

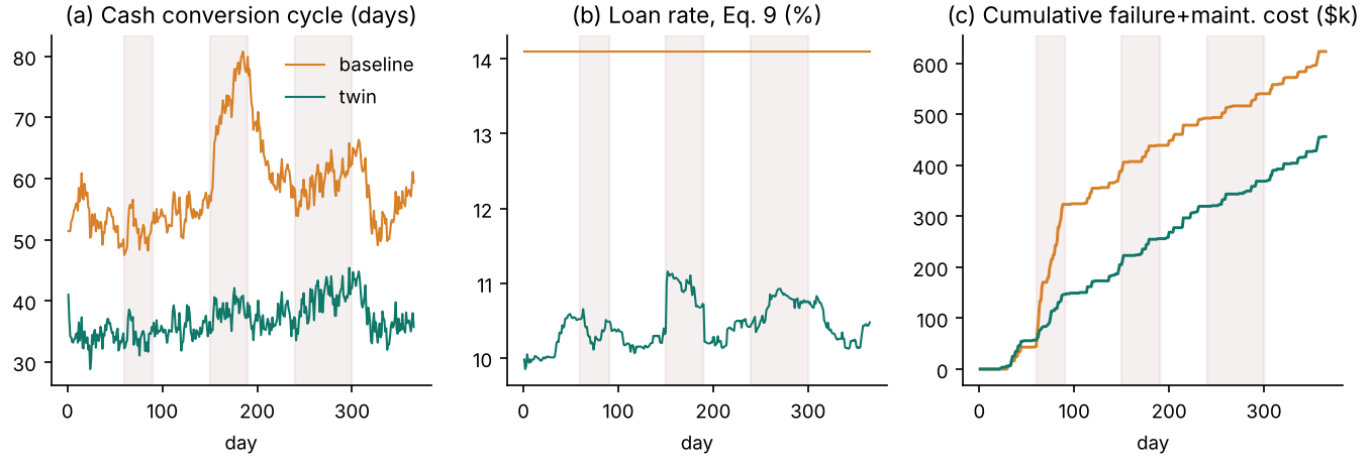


Fig. 4. One stressed year (seed 11, shocks shaded): (a) CCC, (b) loan rate from Eq. 9, (c) cumulative failure and maintenance cost.

E3 — Right-sizing the twin (Gap 2) (Table V, Fig. 5). With the survey's benefits, Standard is recommended, and High-fidelity is a decision reversal: it is accepted statically but rejected under risk. With the measured benefits, Standard also reverses, and only Lite stays positive. Its 18.4% chance of loss exceeds the 10% limit, so FinTwin advises a stage-gated Lite pilot. The present worth of overheads rises from $23k (Lite) to $341k (High), and CO_2 from 0.8 to 67 t a year.

TABLE V. TIERS UNDER SURVEY VS. MEASURED BENEFITS ($K)

Tier	Static (survey)	Risk-adj. (survey)	P(loss)	Static (meas.)	Risk-adj. (meas.)	P(loss)
Lite	370.7	258.2	0.0%	97.4	52.6	18.4%
Standard	684.9	375.4	0.3%	73.6	−73.3	71.2%
High	243.3	−273.3	92.5%	−485.3	−731.5	100%

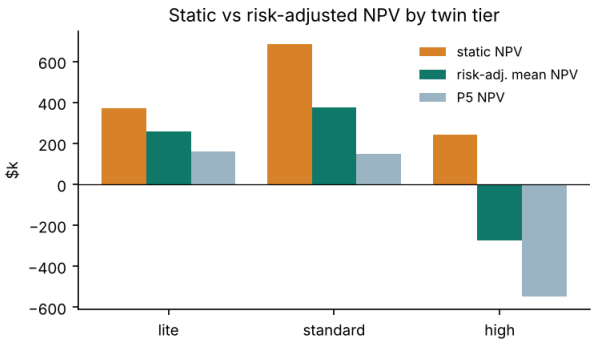


Fig. 5. Static vs. risk-adjusted NPV by tier (survey-assumed benefits).

E4 — ML models (Table VI). The failure model was evaluated on a 25% stratified hold-out of AI4I. The alarm threshold was tuned on out-of-fold training predictions, with a missed failure costing $25k and a false alarm $1.5k. The final model was selected by economic saving rather than AUC.

The DSO model, trained on the 6,000 synthetic invoices, has a mean absolute error of 2.90 days and R^2 of 0.935. A static ERP rule (terms plus past delay) gives 5.84 days and 0.742. Its counterfactual effect of twin-verified delivery is −6.3 days per invoice.

TABLE VI. FAILURE-MODEL COMPARISON (AI4I HOLD-OUT)

Model	ROC-AUC	PR-AUC	Recall @ thr.	Saving vs. run-to-failure
Logistic regr.	0.940	0.502	0.812	65.8%
Random forest	0.972	0.845	0.788	78.1%
HistGB (sel.)	0.963	0.835	0.835	**80.8%**

E5 — Maintenance policy (Fig. 6). In steady state, the best fixed intervals (160–200 min) cost $412k a year and the twin $407k, without any tuning. The worst fixed interval (240 min) costs $628k. Under two overload spells, every fixed interval costs $869k–$1,032k, against $516k for the twin.

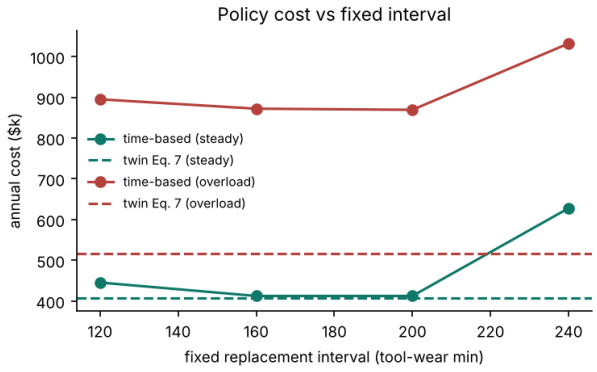


Fig. 6. Annual cost of fixed-interval replacement vs. the twin's Eq. 7 rule.

E6 — Performance (one CPU core):

- a 365-day dual-world twin with three shocks: 2.2 s;
- 1,000 Monte-Carlo runs: 0.005 s (standard error $4.8k);
- 5,000 runs: 0.019 s ($2.1k);
- 20,000 runs: 0.073 s ($1.1k);
- a full three-tier comparison: 0.05 s.

VII. Discussion and Limitations

The central finding is that *a twin should be sized to the disruption risk it hedges, not to the technology available.* Three lessons follow:

1. Static NPV hides the timing of a twin's value: benefits ramp up while costs (refresh, compliance) arrive on schedule.
2. Disruption-driven cash flows are volatile. Through Eq. 5, that volatility raises the hurdle rate from 10% to 13.5% (Standard) and 25% (High, measured benefits).
3. IRR and NPV can disagree: Lite has the highest IRR (62%), while Standard has the highest NPV. FinTwin decides on risk-adjusted NPV under a loss constraint.

The study has these limitations:

- The receivables ledger is synthetic.
- AI4I rows are independent snapshots, not run-to-failure paths, so the daily hazard is an approximation. The calibrated, wear-monotone probabilities partly offset this.
- Tier costs and capture factors are engineering estimates; the tornado and break-even outputs quantify how much they matter.
- Smart-contract factoring and blockchain [A11] are represented only through their effects on DSO and credit.

All parameters are editable in the user interface, so a firm can replace every estimate with its own data.

VIII. Deployment, Repository and User Manual

FinTwin is deployed on Render's free tier at https://fintwin-0ae4.onrender.com; the first visit after idling takes 30–50 s. The `render.yaml` blueprint installs the requirements, regenerates the invoice ledger and retrains both models, so the models always match the server's scikit-learn version. Gunicorn then starts with 2 workers, and `/api/health` serves as the health check.

The repository https://github.com/shreyank26pal-dev/eefm_project contains:

- `frontend/` (dashboard), `backend/` (API, economics engine, twin, ML, experiments) and `dataset/` (AI4I 2020 and the invoice generator);
- 9 unit tests that replicate the survey case;
- `README.md`, with local setup steps;
- `docs/USER_MANUAL.md`: a guide to each module, its inputs and outputs, the REST API and troubleshooting.

IX. Conclusion and Future Work

FinTwin carries the CIE-1 survey from a deterministic case study to a deployed, AI-enabled decision-support system. It reproduces the survey NPV of $684,814. Pricing risk through $r(\sigma)$ and the twin's own overheads lowers it to $375k. Replacing assumed benefits with measured ones changes the recommendation from a $450k Standard twin to a stage-gated $150k Lite pilot.

The project puts these Engineering Economics concepts into practice:

- time value of money: PW, FW, AW, capital recovery and rate of return;
- payback, break-even and depreciation;
- replacement analysis, using EUAC and marginal cost;
- risk analysis, using Monte Carlo, CVaR and a risk-adjusted rate;
- working-capital ratio dynamics: DIO, DSO, DPO and CCC.

Future work:

- calibrating the twin on a partner firm's ERP and sensor logs;
- real options to expand or abandon after the pilot, via the Bellman form of Eq. 4;
- explicit factoring smart contracts, including their fees;
- multi-firm credit sharing.

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